

Justin A. Alvaran

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Professional Summary

Junior Software Engineer and AI/ML Engineer with early-career experience building production web applications and machine learning services. Delivered four Next.js applications with page-load performance improvements of up to 40% and developed an XGBoost model achieving 85% predictive performance on construction cost data. TESDA National Skills Competition 2026 Gold Medalist and Philippine representative at the WorldSkills ASEAN International Competition.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, Java, HTML, CSS, Tailwind CSS
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis
- **Frameworks:** React, Next.js, FastAPI, Django, Express.js
- **AI/ML:** PyTorch, TensorFlow, scikit-learn, XGBoost, Polars, Pandas, NumPy, Kubeflow
- **Generative AI/Automation:** LangChain, LangGraph, CrewAI, LlamaIndex, RAG, n8n, GoHighLevel (GHL)
- **Cloud & DevOps:** AWS, Azure, Docker, GitHub Actions, GitLab CI/CD, Git, MLflow, LangSmith

PROFESSIONAL EXPERIENCE

Cornerstone International Technologies (CITC)

Aug 2024 - Sep 2026

Junior Software Engineer

Georgia, USA (Remote)

- Shipped 4 production web applications with Next.js 15, TypeScript, and Tailwind CSS, translating Figma designs into reusable UI components and improving page-load performance by 30–40% through code splitting and image optimization.
- Architected GitLab CI/CD pipelines with OIDC authentication to automate secure deployments to AWS S3 and CloudFront across multi-tier environments (dev, staging, production).
- Integrated Directus CMS with Next.js via REST APIs and SSG, implementing incremental static regeneration to automate nightly content refreshes.
- Configured Express.js middleware with Google reCAPTCHA Enterprise to defend public-facing forms against automated bot traffic.

General Contractors

Nov 2025 - May 2026

Junior AI/ML Engineer (Contract)

Atlanta, GA (Remote)

- Developed and trained an XGBoost regression model using Python, achieving $R^2 = 0.85$ on a cleaned dataset of 1,000+ historical construction records.
- Engineered data pipeline scripts with Pandas and NumPy to extract features from materials, labor costs, and project timelines, reducing raw data preprocessing time.
- Deployed the model as a containerized microservice with FastAPI async endpoints on AWS and Azure, securing API inputs via Pydantic schema validation, which improved service reliability and reduced input errors

CIRCA

Aug 2026 - Sep 2026

Forward Deployed Engineer Intern

United States (Remote)

- Built and integrated API systems with TypeScript and Express.js, enabling service-to-service communication and RESTful endpoints, and orchestrated data pipelines using Kubeflow in a production distributed architecture, which improved data processing reliability

Kinetrexa Software Private Limited

Jul 2026 - Aug 2026

AI & Machine Learning Intern

Tamil Nadu, India (Remote)

- Developed end-to-end ML pipelines with NumPy, XGBoost, and Kubeflow, covering data collection, preprocessing, model training, hyperparameter tuning, and performance evaluation; used MLflow for version-controlled experiments, which streamlined iteration and improved model performance

Brinicle AI

Jun 2026 - Jul 2026

AI/ML Engineering Intern

United States (Remote)

- Built and evaluated classification and regression models with Python, scikit-learn, and TensorFlow/Keras, creating data-preprocessing pipelines, feature-engineering steps, and model-evaluation metrics, which improved predictive performance on validation data

BookMI

Growth Intern (Technology)

Jan 2026 - Mar 2026

Maharashtra, India (Remote)

- Contributed to product development workflows by writing and debugging code, documenting features, and testing using Git, GitLab CI/CD, and PostgreSQL within agile sprint cycles, which helped shorten release cycles and improve overall code quality

PROJECTS

End-to-End Flood Classification & Edge Deployment System

TESDA National Skills Competition 2026 - Gold Medal Project

- Engineered a competition-grade AI pipeline in 3 days: preprocessed 2,000+ hydrological sensor records with 7% minority class imbalance, applied warm-start transfer learning on an MLP neural network with zero data-leakage validation, achieving 95.76% accuracy and 0.9411 Macro F1 on a 495-sample held-out test set.
- Exported model to ONNX, compiled to OpenVINO IR, achieved $\geq 99\%$ inference consistency, and deployed a standalone real-time alert engine with input validation gates - all with zero dependency on the original training framework.

Loop Engineer CI/CD

- Built an event-driven CI/CD workflow automation service processing failed GitHub Actions runs via signed webhooks to generate evidence-backed repair plans, reducing build failure triage time by 40%.
- Developed an Engineering Memory Engine matching build failures with historical resolutions, identifying failure classifications and likely code owners with 94% confidence.
- Implemented policy-gated routing to deliver repair plans with confidence/risk scores to Slack, Teams, and email, and automatically open Jira tickets.

AI Water Meter

- Built a Chrome/Edge Manifest V3 extension with local-first architecture estimating carbon, water, and energy footprints of LLM sessions in real-time with $< 50\text{ms}$ tracking latency.
- Engineered a local token estimation engine parsing prompts client-side, translating token counts into environmental resource metrics based on published empirical research.
- Built a React/Vite dashboard and leaderboard with Supabase backend and a companion VS Code extension for editor-text estimation and Claude Code terminal log syncing.

UrbanGrid

- Trained and evaluated Random Forest, XGBoost, and SVM models using 5-fold cross-validation and GridSearchCV hyperparameter tuning to forecast energy-demand requirements.
- Trained a ResNet50 CNN on 50,000 infrastructure images achieving 98% validation accuracy with Grad-CAM focus mapping; served via FastAPI async endpoints with Pydantic validation.
- Created a Streamlit monitoring dashboard with real-time /health checks, containerized via Docker.

Achievements

- **TESDA National Skills Competition 2026 - Artificial Intelligence (Skill 53):** Philippines | Sep 2026; Gold Medalist & National Champion | ASEAN International Competition Candidate; Awarded 1st Place nationally in the WorldSkills AI discipline, competing under timed, pressure-tested conditions judged by industry experts. Selected to represent the Philippines at the WorldSkills ASEAN Competition.

Certifications

- Microsoft Certified: Azure AI Engineer Associate (AI-102): Microsoft
- Generative AI for Executives: Amazon Web Services
- AI Fluency: AI Capabilities & Limitations: Anthropic
- Claude Code 101: Anthropic

EDUCATION

University of Immaculate Conception

BS, Computer Science

2023 - 2026

Davao, Philippines